KEYPAD-BASED PRODUCT COUNTING AND AUTOMATIC BATCH PACKING SYSTEM WITH STATION SENSING USING ARM7 LPC2129

A MINI PROJECT REPORT

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1. ABSTRACT

This project presents a keypad-based product counting and automatic batch packing system using the ARM7 LPC2129 microcontroller. The user enters the required batch quantity through a 4×4 matrix keypad and starts the conveyor system. Two IR sensors detect and monitor product movement at different points. The LPC2129 processes the sensor signals, updates the product count and controls the conveyor motor. A 16×2 LCD displays the count and system status. Red and green LEDs indicate conveyor operation and batch completion, while a buzzer provides an alert. The project demonstrates practical use of Embedded C, GPIO interfacing, external interrupts, sensor interfacing and motor control.

2. INTRODUCTION

Product counting and batch preparation are common operations in manufacturing and packaging industries. Manual counting can require time and may lead to errors. An embedded automation system can help make the process more consistent.

In this project, the ARM7 LPC2129 acts as the main controller. The keypad is used to enter the required quantity, and two IR sensors monitor products moving along the conveyor. Based on the sensor inputs, the controller updates the count and operates the motor. The LCD, LEDs and buzzer provide information about the system condition.

3. OBJECTIVES OF THE PROJECT

- Enter the required product quantity using a matrix keypad.
- Detect and monitor product movement using two IR sensors.
- Count products automatically using the LPC2129 microcontroller.
- Control a conveyor motor according to system operation.
- Display product count and system status on a 16×2 LCD.
- Indicate conveyor operation and batch completion using LEDs and a buzzer.

4. COMPONENTS REQUIRED

S. No.	Component	Purpose
1	ARM7 LPC2129	Main controller
2	IR Sensor 1	Detects incoming products
3	IR Sensor 2	Monitors product movement at the station
4	4×4 Matrix Keypad	Enters quantity and system commands
5	16×2 LCD	Displays count and system status
6	DC Motor and Motor Driver	Drives the conveyor belt
7	Red and Green LEDs	Indicate operation and batch completion
8	Buzzer	Provides an audible alert
9	Regulated Power Supply	Powers the controller and modules
10	Connecting Wires	Electrical connections

5. HARDWARE DESCRIPTION

5.1 ARM7 LPC2129

The LPC2129 is the main controller. It reads keypad and IR sensor inputs, processes the counting logic, controls the motor and updates the LCD, LEDs and buzzer.

5.2 IR Sensors

An IR sensor module uses an infrared transmitter and receiver to detect an object. IR Sensor 1 is used for product detection and is configured to generate an external interrupt. IR Sensor 2 monitors the product at the station. Sensor output polarity depends on the module.

5.3 4×4 Matrix Keypad

The matrix keypad provides the user interface for entering the required product quantity and selecting commands such as start or clear.

5.4 16×2 LCD Display

The LCD displays the entered quantity, current product count and messages such as system ready, conveyor running, fault or packing complete.

5.5 DC Motor and Motor Driver

The DC motor drives the conveyor belt. A motor driver interfaces the motor with the microcontroller because the motor requires more current than a GPIO pin can supply.

5.6 LEDs and Buzzer

The red LED indicates conveyor operation and the green LED indicates batch completion. The buzzer provides an audible notification when the batch is completed or an alert is required.

6. BLOCK DIAGRAM

4×4 Keypad	**IR Sensor 1** **Product detection**	**IR Sensor 2** **Station sensing**
	ARM7 LPC2129 **Main Controller**	
16×2 LCD	**Motor Driver → DC Motor** **Conveyor**	**LEDs + Buzzer**

Figure 1: Functional block representation of the product counting and batch packing system.

7. CIRCUIT INTERFACING

The keypad, IR sensors, LCD, motor driver, LEDs and buzzer are interfaced with the LPC2129 through suitable GPIO and peripheral pins. IR Sensor 1 is connected to EINT0 so product detection can be handled using an external interrupt. The motor is connected through a suitable motor driver. The motor supply and controller supply must be wired according to their ratings, with a common ground where required.

Signal / Device	LPC2129 connection
LCD RS / EN	P0.12 / P0.13
LCD data D4–D7	P0.22–P0.25
Keypad columns	P1.23–P1.26
Keypad rows	P1.27–P1.30
IR Sensor 1 / EINT0	P0.16
IR Sensor 2	P0.18
Red LED / Green LED	P0.29 / P0.30
Motor driver inputs	P0.8 / P0.9
Buzzer	P0.14

Note: Verify these pin assignments against the final hardware wiring and tested source code before submission. Use a suitable driver circuit for the motor. Do not connect a motor directly to a microcontroller GPIO pin.

8. WORKING PRINCIPLE

1. The system is powered ON and the LCD displays the initial status.
2. The user enters the target product quantity using the keypad.
3. After the START command, the conveyor motor runs and the red LED indicates operation.
4. When a product is detected by IR Sensor 1, the external interrupt signals the controller.
5. IR Sensor 2 monitors the product at the station, and the controller updates the counting process.
6. The LCD displays the current count and system status.

7. When the required quantity is reached, the conveyor stops and batch completion is indicated by the green LED and buzzer.

9. SOFTWARE REQUIREMENTS

- Embedded C programming language.
- Keil µVision IDE for writing and compiling firmware.
- LPC21xx device header file for register access.
- LCD driver routines for displaying messages and counts.
- Keypad scanning routine for reading user input.
- Proteus Design Suite for circuit simulation, where applicable.

10. PROGRAM / SOURCE CODE

The firmware is developed in Embedded C. The program configures the GPIO pins, initializes the LCD and keypad, configures the external interrupt for IR Sensor 1, and executes the product-counting and conveyor-control sequence.

- Initialize the required GPIO pins and peripherals.
- Initialize the LCD and keypad.
- Read the target quantity entered by the user.
- Wait for the START command.
- Start the conveyor and monitor the IR sensor signals.
- Update the count and LCD status based on sensor events.
- Stop the conveyor and indicate completion when the target is reached.

Add the final tested main program, interrupt routine, LCD header and keypad header files to the project repository. Confirm that the documented pin assignments match the final code.

11. APPLICATIONS

- Product counting in manufacturing industries.
- Batch preparation in packaging industries.
- Counting bottles, packets or small components on conveyor systems.
- Food-processing and pharmaceutical packaging lines.
- Educational demonstrations of embedded automation.

12. ADVANTAGES

- Reduces manual counting effort.
- Helps reduce human counting errors.
- Provides clear visual and audible system indications.
- Combines sensor interfacing, interrupts and motor control in one application.
- Can be extended for industrial monitoring and data logging.

13. CONCLUSION

The project demonstrates an embedded solution for automatic product counting and batch packing using the ARM7 LPC2129 microcontroller. By combining a keypad, two IR sensors, an LCD, a conveyor motor, LEDs and a buzzer, the system performs quantity-based product handling with clear status indications. The project provides practical experience in Embedded C, GPIO interfacing, external interrupts, sensor interfacing and motor control. It can be further developed with production data storage, wireless monitoring and IoT-based supervision.

PROJECT DETAILS

Project	Keypad-Based Product Counting and Automatic Batch Packing System with Station Sensing
Microcontroller	ARM7 LPC2129
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